Experiment 7

Question

Capture video from a Web Camera and display the video in slow motion and in fast motion.

Aim

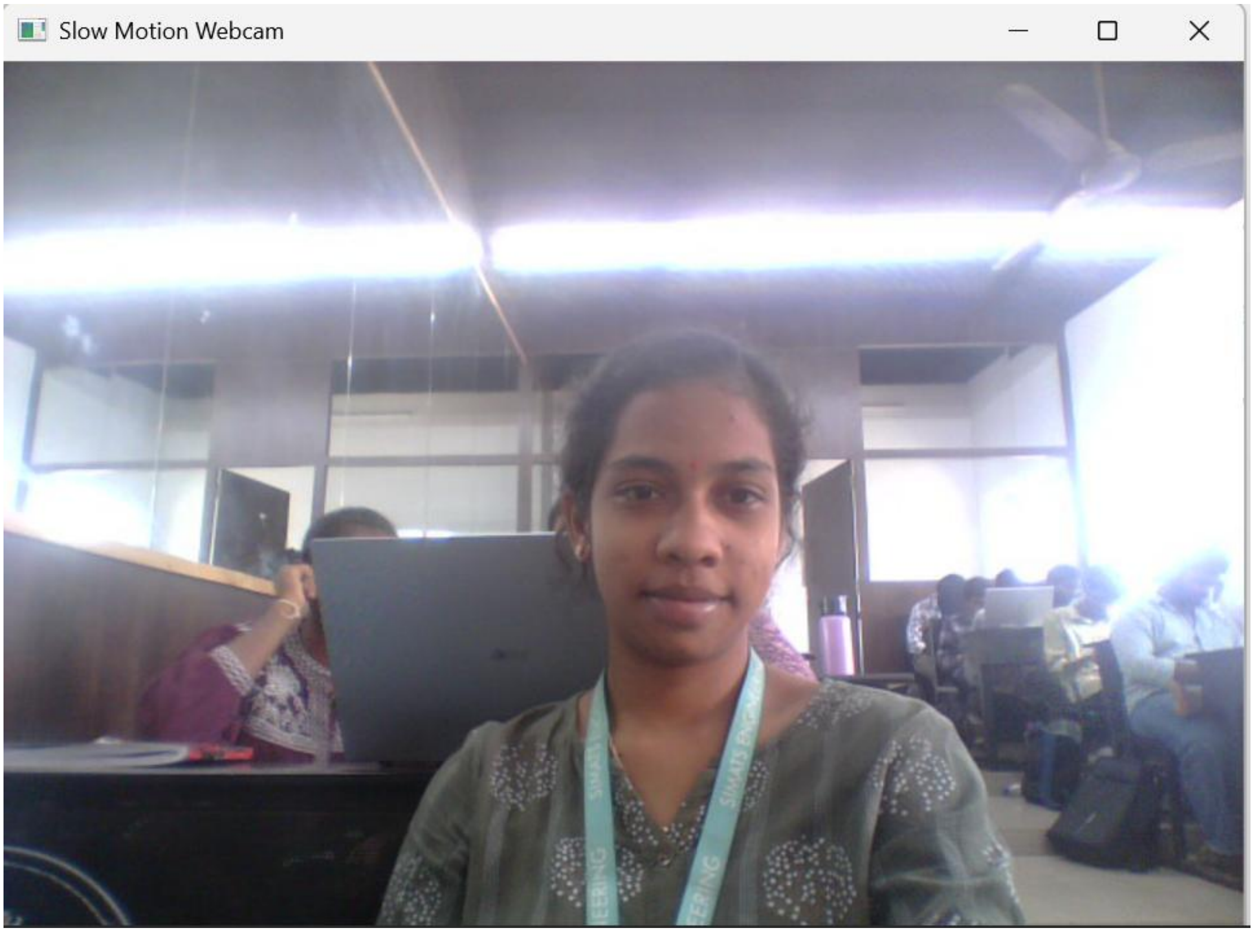
To capture live video from a webcam using Python and OpenCV and display it in slow motion and fast motion.

Procedure

1. Import the OpenCV (cv2) library.
2. Open the webcam using cv2.VideoCapture(0).
3. Check whether the webcam is opened successfully.
4. Capture the video frame by frame.
5. Resize each frame for better display.
6. Display the video.
7. Use cv2.waitKey(100) for slow motion & cv2.waitKey(10) for fast motion.
8. Press **'q'** to stop video and save output video

Code (Slow Motion)

```
import cv2


# Open webcam
camera = cv2.VideoCapture(0)
# Check webcam
if not camera.isOpened():
    print("Webcam not found!")
else:
    while True:
        ret, frame = camera.read()
        if not ret:
            break


        # Resize the frame
        frame = cv2.resize(frame, (640, 480))


        # Display webcam
        cv2.imshow("Slow Motion Webcam", frame)


        # Slow motion
        if cv2.waitKey(100) & 0xFF == ord('q'):
            break


camera.release()
cv2.destroyAllWindows()
```


Slow Motion Webcam

Code (Fast Motion)

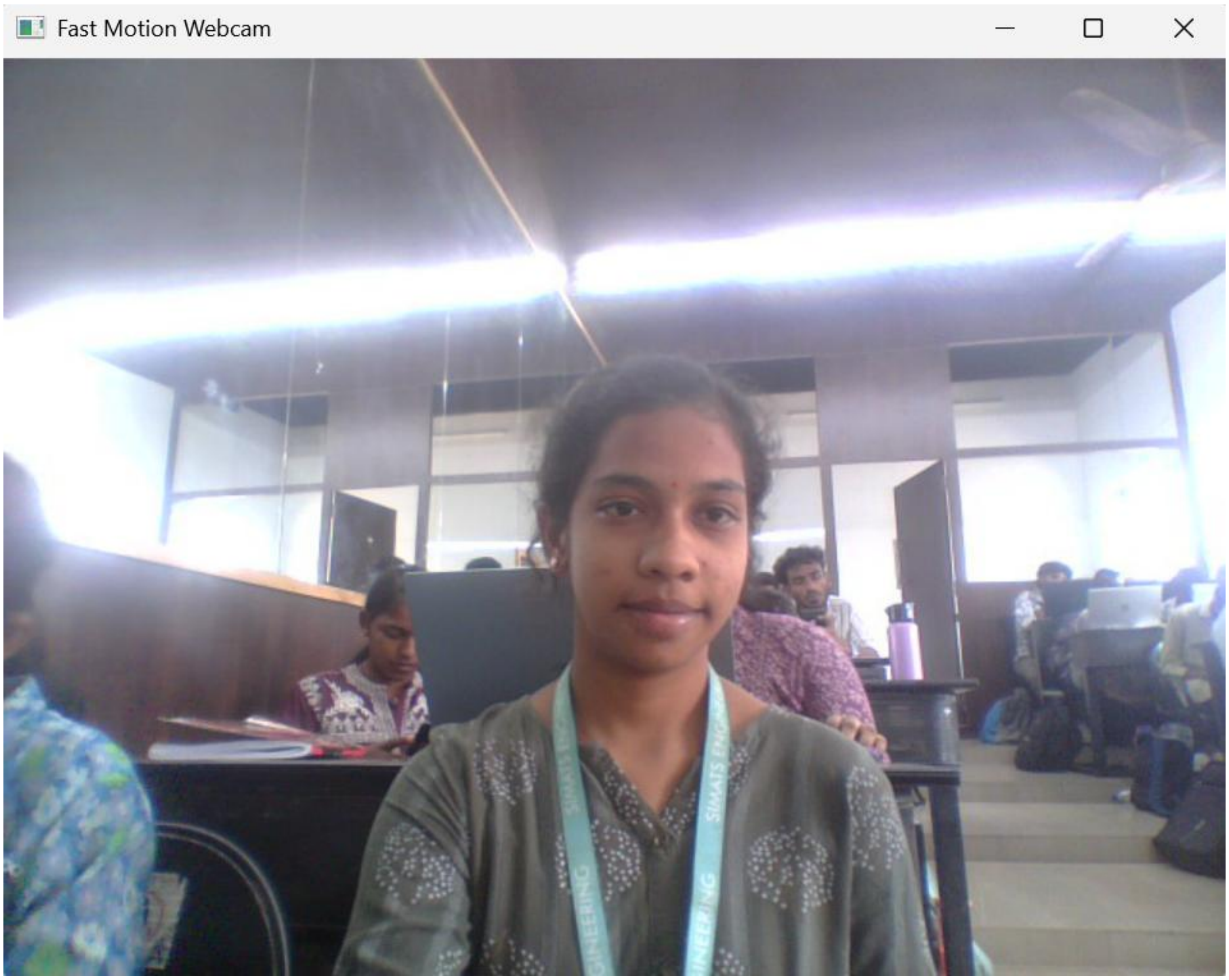
```
import cv2


# Open webcam
camera = cv2.VideoCapture(0)
# Check webcam
if not camera.isOpened():
    print("Webcam not found!")
else:
    while True:
        ret, frame = camera.read()


        if not ret:
            break
        # Resize the frame
        frame = cv2.resize(frame, (640, 480))


        # Display webcam
        cv2.imshow("Fast Motion Webcam", frame)


        # Fast motion
        if cv2.waitKey(10) & 0xFF == ord('q'):
            break


camera.release()
cv2.destroyAllWindows()
```

Input

Input Device: Laptop Webcam

Output

1. **Slow Motion Webcam** : Displays the live webcam video at a slower speed.
2. **Fast Motion Webcam** :Displays the live webcam video at a faster speed.



Result

The live video was successfully captured from the webcam using OpenCV and displayed in both slow motion and fast motion.